FYP22 Monthly Progress Report: September 2026

Group: 20	Project: A multi-RF head receiver array for AI-native sensing and communications

Overall Monthly Summary

During this month we focused on developing the key technical foundations required for the proposed AI-native ISAC receiver system. Work included studying OFDM and OTFS waveforms, communication and sensing signal processing, and identifying suitable algorithms for UAV sensing and detection. We also investigated important system design parameters such as SNR, bandwidth, sensing range, transmit power and link budget requirements. Simulation work was carried out to understand waveform transmission, channel effects, receiver processing, and the extraction of sensing parameters such as delay and Doppler. In parallel, research papers on OTFS-based UAV detection were reviewed to evaluate existing methods, their feasibility and limitations. Antenna development also progressed through the design and simulation of a 4×4 microstrip patch array for the 5.7–6.0 GHz band, including feed-network design, impedance matching and HFSS model optimization. Overall, the month established a stronger theoretical and simulation foundation for the next stages of RF, sensing and communication system development.

Index no.: 220692R	Name: Weragoda W.A.A.P
In this month our first goal was to find a method to start with the RF (Radio Frequency) chain design. For that first we decided to understand what kind of algorithms we can implement in order to successfully implement the ISAC on our receiver and what are the expected parameter values that receiver should have to support those algorithms. In there we figured out that SNR(Signal to Noise Ratio) is critical for sensing application. So we carried out calculations to figure out such parameters and concluded design parameters like instantaneous bandwidth , sensing range , required SNR at the receiver and required transmit power. Then I also involved in some parts to simulate one or more selected algorithms that we can implement in our design. Also carried out weekly meetings with the supervisors.	
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Index no.: 220036L	Name: Anuradha D.M.B.P.
During this month, significant progress was made in understanding the OFDM waveform and its application in an ISAC system. The OFDM transmission and reception process was studied, including how the waveform is generated at the transmitter, how it propagates through the communication channel, and how the received signal is processed to recover the transmitted information. The signal processing techniques used to extract target-related parameters, such as time delay and Doppler shift, from the received waveform were also investigated. In addition, the radar and communication link budget calculations were carried out to evaluate the expected system performance under different operating conditions. Work was also conducted on identifying and evaluating suitable waveforms for the proposed ISAC implementation, considering their suitability for both communication and sensing functions. These studies provided a better understanding of the requirements and design considerations for the overall ISAC system.	
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Index no.: 220511N	Name: Ranaweera R.P.D.D.H.
I designed a 4×4 microstrip patch array antenna for the 5.7–6.0 GHz band (centred at 5.85 GHz) on a 1.6 mm FR-4 substrate. The 16 patches are fed through a 1 → 16 corporate feed network that uses 70.7 quarter-wave transformers to keep each branch matched to 50 , with equal-length paths so all patches are fed in phase. I wrote a Python script that automatically builds the full parameterised HFSS model (geometry, materials, 50 lumped port, radiation boundary, frequency sweeps and gain reports) and checks the geometry for continuity, clearance and symmetry. Using the simulation results, I rescaled the patch length to tune the resonance to 5.85 GHz, added an open-circuit matching stub at the input to improve the match across the band, replaced the rounded junction pads with flat T-junctions, and adapted the model to solve within the mesh limit of the HFSS Student licence.	
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Index no.: 220452H	Name: Peiris P.I.U.
In this month, I mainly focused on developing my understanding of the communication and sensing techniques related to our ISAC project. I first studied the basic theory of OFDM to understand its modulation, signal transmission and receiver processing. I also participated in the simulation work to gain a better understanding of how the transmitted signal is affected by the channel and recovered at the receiver. In addition, I started studying research papers related to OTFS-based ISAC systems for UAV detection. Through this, I am currently identifying how OTFS is used for UAV sensing, the signal processing methods used for detection, the feasibility of these approaches and their main limitations. This work will help us select suitable methods and simulation models for the next stage of the project.	
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Supervisor's Certification

Supervisor's Name	**Signature**	**Date**
Dr. Chamira U. S. Edussooriya		30 September 2026